

# M13.8 Instruments 345-406 — Quiz

40 questions covering every subject in the batch · Level 1: 22 · Level 2: 15 · Level 3: 3

Part-66 format: three answers, one correct, no negative marking. Pass mark 75% — 30 of 40.

At the Part-66 allowance of 75 seconds a question, 40 questions is 50 minutes.

1. What does the abbreviation EICAS stand for
  - a. Engine Indicating and Centralized Alerting System
  - b. Engine Indicating and Crew Alerting System
  - c. Electronic Indicating and Crew Alerting System
2. Why do the aircraft monitoring systems of the modern cockpit carry different names such as EICAS or ECAM
  - a. Because the name depends on the aircraft type
  - b. Because the name depends on the display size
  - c. Because the name depends on the flight phase
3. Which task do EICAS and ECAM perform in the modern cockpit
  - a. They perform maintenance data recording
  - b. They perform aircraft monitoring
  - c. They perform flight monitoring
4. Why is the CMS counted as part of aircraft monitoring
  - a. Because it delivers information to the flight crew
  - b. Because it delivers information to the maintenance personnel
  - c. Because it delivers information to the standby instruments
5. The electronic displays of a glass cockpit are lost in flight. Which speed and attitude information remains available from the standby set?
  - a. Airspeed from the standby airspeed indicator and attitude from the standby horizon
  - b. Mach number from the standby machmeter and attitude from the standby horizon
  - c. Airspeed from the standby airspeed indicator and heading from the standby horizon
6. What is provided to each pilot for flight attitude and flight monitoring
  - a. His own set of engine and systems monitoring instruments
  - b. His own set of flight attitude and flight monitoring instruments
  - c. One set of flight attitude and flight monitoring instruments shared by both pilots
7. What does the Basic T set of flight monitoring instruments tell the pilot about
  - a. Aircraft attitude, airspeed and altitude
  - b. Aircraft airspeed, altitude and fuel quantity
  - c. Aircraft attitude, heading and engine thrust
8. Where is the magnetic standby compass often found
  - a. Between the windscreens
  - b. Beside the altimeter
  - c. Below the airspeed indicator
9. In which layer of the atmosphere do the numerous weather activities take place
  - a. In the stratosphere
  - b. In the thermosphere
  - c. In the troposphere
10. What is the name of the boundary between the troposphere and the stratosphere
  - a. The tropopause
  - b. The stratopause
  - c. The mesopause
11. What is the other name for dynamic pressure
  - a. Ram pressure
  - b. Atmospheric pressure
  - c. Barometric pressure
12. What is the other name for static pressure
  - a. Ram pressure
  - b. Atmospheric pressure
  - c. Total pressure

13. What is the standard atmosphere sea level pressure in the imperial system
- 29.92 inHg
  - 14.7 inHg
  - 29.92 mmHg
14. How does the air temperature of the standard atmosphere behave with increasing height directly above the tropopause
- It stops falling and stays constant
  - It continues to fall at the same rate
  - It begins to rise with height
15. What is the altitude of the tropopause in the international standard atmosphere
- 20 km
  - 11 km
  - 8 km
16. Which air data parameter is presented on the left-hand scale of the PFD
- Airspeed
  - Altitude
  - Vertical speed
17. What is set on the altimeter when QNH is used
- The current air pressure at mean sea level
  - The standard pressure of 1013.2 hPa
  - The current air pressure at the airfield
18. What is a flight level
- The height above the airfield pressure level
  - The height above the mean sea level pressure of the day
  - The height above the 1013.25 hPa pressure level
19. What is the transition altitude
- After take-off, the height at which the altimeter is set from STD to QNH
  - Prior to landing, the height at which the altimeter is set from QNH to STD
  - After take-off, the height at which the altimeter is set from QNH to STD
20. What is the purpose of reducing the vertical separation minimum
- To allow aircraft to climb above the tropopause
  - To reduce the workload of the air data computers
  - To accommodate more aircraft in the same airspace
21. What is RVSM airspace
- Airspace above FL 285 with 500 ft vertical separation
  - Airspace above FL 290 with 2000 ft vertical separation
  - Airspace above FL 290 with 1000 ft vertical separation
22. An aircraft is being equipped for flight in RVSM airspace. How many independent altimeters must it have?
- 2
  - 1
  - 3
23. What does the abbreviation IAS stand for
- Inertial airspeed
  - Increased airspeed
  - Indicated airspeed
24. What is the equivalent airspeed
- The calibrated airspeed after the compressibility error has been removed
  - The indicated airspeed after the instrument error has been removed
  - The true airspeed after the density calculation has been removed
25. For which purpose is the true airspeed needed
- For navigation purposes
  - For engine trim purposes
  - For design data purposes
26. Why is the equivalent airspeed lower than the calibrated airspeed at high speed and altitude?
- Because reduced density lowers the total pressure, so the indication is too high
  - Because compression of the air lowers the static pressure, so the indication is too low
  - Because compression of the air raises the total pressure, so the indication is too high

27. The compressibility error for a flight condition is known in knots. What is done with this value to obtain the equivalent airspeed?
  - a. It is subtracted from the true airspeed
  - b. It is subtracted from the calibrated airspeed
  - c. It is added to the calibrated airspeed
28. What is the residual pressure inside an aneroid capsule
  - a. About 5 mbar
  - b. About 50 mbar
  - c. About 1013 mbar
29. A Bourdon tube transducer is fitted with a synchro transmitter. Which quantity does this sensor convert into an electrical signal?
  - a. The temperature of the tube wall
  - b. The position of the tube tip
  - c. The pressure inside the tube
30. What is the amount of tube tip movement proportional to
  - a. The signal frequency
  - b. The tube temperature
  - c. The applied pressure
31. Which temperature range must a pressure sensor flight instrument cover
  - a. -50 °C to +70 °C
  - b. -40 °C to +55 °C
  - c. -70 °C to +50 °C
32. An instrument panel is supplied only from the aircraft pitot and static system. Which group of instruments is this?
  - a. Airspeed indicator, altimeter, variometer and radio altimeter
  - b. Airspeed indicator, altimeter, machmeter and attitude indicator
  - c. Airspeed indicator, altimeter, machmeter and variometer
33. Which effect does friction have in the transfer mechanism of an aneroid instrument
  - a. It raises the capsule hysteresis
  - b. It falsifies the measured result
  - c. It damps the capsule stroke
34. What is meant by the hysteresis of a membrane capsule
  - a. The curve with increasing pressure differs from the curve with increasing temperature
  - b. The curve with increasing pressure differs from the curve with decreasing pressure
  - c. The stroke lags behind the pressure when a constant pressure acts for a long time
35. What causes the elastic aftereffect of a capsule
  - a. A temperature change acting on the capsule mounting
  - b. A rapidly changing pressure acting on the capsule during a climb
  - c. A constant pressure acting on the capsule over a longer period of time
36. The hysteresis and the elastic aftereffect of a capsule are shown as characteristic curves. Which quantity of the capsule is plotted against the applied pressure?
  - a. The needle friction
  - b. The diaphragm stroke
  - c. The residual capsule pressure
37. What is normally inside the reference pressure chamber of a vibrating transducer
  - a. Usually pitot air
  - b. Usually a vacuum
  - c. Usually ambient air
38. In which circuit are the strain gauge resistances connected
  - a. In a resolver winding
  - b. In a series chain
  - c. In a Wheatstone bridge
39. How many strain gauges are normally fitted on the diaphragm of a piezoresistive transducer
  - a. Two resistors
  - b. Four resistors
  - c. Eight resistors
40. No input pressure is applied to a piezoresistive transducer. What is the condition of the base plate and the bridge?
  - a. The plate is straight, the front resistors are lower and the bridge is balanced
  - b. The plate is bent, all resistors are equal and the bridge is balanced
  - c. The plate is straight, all resistors are equal and the bridge is balanced

## Answer key

| No. | Ans | Level | LTT page | Why                                                                                                                               |
|-----|-----|-------|----------|-----------------------------------------------------------------------------------------------------------------------------------|
| 1   | b   | L1    | 2        | EICAS is given as Engine Indicating and Crew Alerting System.                                                                     |
| 2   | a   | L2    | 2        | The text states that aircraft monitoring instruments have different names depending on aircraft type.                             |
| 3   | b   | L2    | 2        | EICAS and ECAM are the names given to the aircraft monitoring instruments.                                                        |
| 4   | b   | L2    | 2        | The CMS is used for aircraft monitoring because it delivers information to maintenance personnel.                                 |
| 5   | a   | L3    | 2        | The standby set contains an airspeed indicator and a horizon, but no machmeter.                                                   |
| 6   | b   | L1    | 8        | The manual states the instrumentation is duplicated so that each pilot has his own set.                                           |
| 7   | a   | L1    | 14       | Those three parameters are the ones the Basic T set is defined to present.                                                        |
| 8   | a   | L1    | 16       | Its usual mounting position is between the windscreens, clear of the main panel.                                                  |
| 9   | c   | L1    | 20       | Weather is contained in the troposphere.                                                                                          |
| 10  | a   | L1    | 21       | The tropopause caps the troposphere.                                                                                              |
| 11  | a   | L1    | 22       | Dynamic pressure is the ram pressure produced by the aircraft's motion.                                                           |
| 12  | b   | L1    | 22       | Static pressure is the atmospheric pressure surrounding the aircraft.                                                             |
| 13  | a   | L1    | 24       | 1013.25 hPa is the same pressure as 29.92 inches of mercury.                                                                      |
| 14  | a   | L3    | 24       | Identical temperatures at both levels show the lapse rate has stopped at the tropopause.                                          |
| 15  | b   | L1    | 25       | The chart places the tropopause at 11 km.                                                                                         |
| 16  | a   | L1    | 29       | Airspeed occupies the left-hand tape of the PFD.                                                                                  |
| 17  | a   | L1    | 30       | QNH is the actual pressure reduced to mean sea level.                                                                             |
| 18  | c   | L1    | 30       | Flight levels are counted from the standard 1013.25 hPa pressure surface.                                                         |
| 19  | c   | L2    | 30       | The transition altitude is the climb-out point for changing to the standard setting.                                              |
| 20  | c   | L2    | 32       | More usable levels mean more aircraft fit into the same volume of airspace.                                                       |
| 21  | c   | L1    | 32       | RVSM airspace starts above FL 290 and uses 1000 ft spacing.                                                                       |
| 22  | a   | L1    | 32       | Two independent altimeters are required.                                                                                          |
| 23  | c   | L1    | 36       | The text defines IAS as indicated airspeed.                                                                                       |
| 24  | a   | L2    | 38       | EAS is what is left when the compressibility error is taken out of CAS.                                                           |
| 25  | a   | L1    | 38       | The manual states TAS is required for navigation.                                                                                 |
| 26  | c   | L3    | 38       | The extra total pressure from compression makes the reading too high, so the correction reduces the speed.                        |
| 27  | b   | L2    | 39       | The 15 knot compressibility error is taken off the calibrated airspeed of 293 knots to give the equivalent airspeed of 278 knots. |
| 28  | a   | L1    | 40       | The capsule keeps a residual pressure of about 5 mbar.                                                                            |
| 29  | b   | L2    | 42       | The sensor senses tip position, not pressure directly.                                                                            |
| 30  | c   | L2    | 44       | Tip travel is proportional to the applied pressure.                                                                               |
| 31  | c   | L1    | 46       | The required range is -70 °C to +50 °C.                                                                                           |
| 32  | c   | L2    | 47       | All four of these are pressure sensing instruments shown on the panel.                                                            |
| 33  | b   | L2    | 48       | Friction in the mechanism corrupts the reading.                                                                                   |
| 34  | b   | L2    | 48       | Hysteresis is the difference between the rising and falling pressure characteristics.                                             |
| 35  | c   | L2    | 48       | Long-held constant pressure makes the capsule yield slowly.                                                                       |
| 36  | b   | L2    | 49       | Both capsule errors are shown as the diaphragm stroke plotted against the applied pressure.                                       |
| 37  | b   | L1    | 52       | The reference chamber usually holds a vacuum.                                                                                     |
| 38  | c   | L1    | 55       | The strain gauges are wired as a Wheatstone bridge.                                                                               |
| 39  | b   | L1    | 56       | Normally four strain gauges are used.                                                                                             |
| 40  | c   | L2    | 56       | With no pressure the plate stays straight, all resistances are equal and the bridge balances.                                     |